Experiment: Student Performance Prediction using Multiple Linear Regression

Aim

To develop a Student Performance Prediction System using the Multiple Linear Regression algorithm to predict final examination marks based on study hours, attendance, internal marks, and assignment scores.

Tools Required

- Python 3.x
- Pandas
- Scikit-learn
- VS Code

Code:

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import r2_score

data = pd.read_csv("E:/New folder/New folder/Student_performance.csv")

X = data.iloc[:, :-1]

y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predicted Marks:", y_pred)

print("R2 Score:", r2_score(y_test, y_pred))
```

Output

: > house price prediction.py > ...

```python
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
data = fetch_california_housing()
X = data.data
y = data.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("R2 Score:", round(r2_score(y_test, y_pred), 2))
```

Ports

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\chandravardhan reddy\AppData\Local\Programs\Microsoft VS Code> & "C:\Users\chandravardhan reddy\AppD
" "e:/house price prediction.py"
R2 Score: 0.58
PS C:\Users\chandravardhan reddy\AppData\Local\Programs\Microsoft VS Code>
```

Result

Multiple Linear Regression model was successfully implemented.